ActionShap: Evaluating Recommendation Explanations Beyond Deletion with Bounded Interventions

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Recommendation explanations are commonly evaluated by deleting interactions and measuring changes in model output, although deployed systems may instead discount or suppress evidence within operational bounds. Deletion faithfulness and intervention validity are therefore distinct. We introduce ActionShap, an offline protocol for auditing recommendation attributions under bounded profile interventions. It separates pointwise attribution–effect alignment, signed directional validity, and budgeted joint decision quality; fixes temporal splits, candidates, tie handling, and distinct-user inference; and evaluates all methods against the same policy and exact budget-two oracle. On MovieLens-1M and Amazon Digital Music, using quality-gated ItemKNN with 1,000 users and five seeds, Monte Carlo Shapley has positive bounded-minus-deletion alignment differences (+0.017 and +0.129). LIME and leave-one-out achieve higher absolute bounded alignment (up to 0.98, versus 0.78 and 0.41 for Shapley) but negative differences. These perturbation effects do not determine decisions: realized NDCG differences among the principal methods are at most 0.003. Shapley and LIME are practically equivalent under the declared ±0.005 margin on both datasets, although LIME is statistically higher on MovieLens. ActionShap thus distinguishes perturbation sensitivity from intervention decision quality and shows why deletion scores alone can mischaracterize explanations under executable interventions.

CCS Concepts: • **Information systems** → **Recommender systems**; • **Computing methodologies** → *Machine learning*; *Machine learning algorithms.*

Additional Key Words and Phrases: explainable recommendation, attribution evaluation, intervention evaluation, counterfactual evaluation, recommendation faithfulness, recommender systems, cooperative game theory

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1 Introduction

An explanation is often consumed as advice about where to act. A user, analyst, or system operator does not usually ask only whether a recommendation score changes when an input is deleted; they ask what will happen if a factor is discounted, suppressed, or otherwise modified within an available operating policy. The distinction is important in recommender systems because historical interactions are evidence, not necessarily controls. Removing an interaction can be a useful faithfulness diagnostic while being an unrealistic intervention.

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Concretely, a recommender engineer can use such an audit to identify historical signals that may be safely discounted, while a platform auditor can test whether an explanation is aligned with changes that the deployed scorer can actually execute. These are system-side diagnostic uses: the protocol does not by itself authorize a profile change or establish that a user wants it.

This paper studies the resulting evaluation gap. We ask whether an attribution ranking predicts the effect of a declared system-executable modification to the interaction history, and whether the explanation leads to a useful joint intervention under a fixed budget. The question is deliberately narrower than recourse: ActionShap does not claim that a user can cause a profile change, does not estimate causal effects in the world, and does not propose a new recommender architecture. It audits an explanation inside a frozen recommender and an explicitly declared intervention simulator.

The protocol is motivated by three observations. First, deletion and bounded modification are different estimands. A deletion moves a factor to a baseline that may be far outside an operational range. Second, pointwise attribution quality does not determine decision quality: a method can rank individual effects well while selecting a harmful pair, or select a good pair despite modest average rank agreement. Third, a method comparison is not interpretable without a chance control, valid-user counts, convergence diagnostics, and inference over distinct users rather than repeated seed–user records.

ActionShap addresses these issues with a history-conditioned interaction-player game. For each user, the most recent training interactions form the player set. A frozen ItemKNN model computes scores from the retained history at inference time, allowing masking and bounded downweighting to change the profile without retraining. The primary characteristic function is a continuous target-margin utility selected by an independent convergence study; NDCG@10 remains the operational ranking outcome. Every method receives the same temporal users, candidates, intervention strengths, action space, and tie rules.

We test two directional hypotheses. **H1** concerns sensitivity to perturbation semantics rather than the superiority of any attribution method: deletion and bounded-weight interventions evaluate distinct response surfaces of the same recommender. Because leave-one-out attribution is algebraically aligned with singleton deletion (Appendix A.2), its deletion correlation is a diagnostic ceiling rather than an empirical advantage. Absolute bounded-intervention alignment is therefore the primary quantity; the bounded-minus-deletion change is reported only as a perturbation-sensitivity diagnostic.

H2 states that pointwise alignment does not determine joint decision quality, so method rankings under AIA and NDCG regret may disagree. This interpretation would be falsified if bounded-minus-deletion changes predicted lower joint-decision regret while absolute bounded alignment added no information. The primary bounded intervention uses $\rho = 0.5$, an interpretable halfway discount that lies strictly between deletion ($\rho = 0$) and the identity ($\rho = 1$); $\rho = 0.25$ is a declared sensitivity to a stronger discount.

Our contributions are fourfold:

(1) We formalize a recommendation-specific evaluation protocol that separates deletion faithfulness, bounded intervention, and budgeted decision quality under a frozen model.
(2) We provide complementary magnitude, signed-direction, success, abstention, regret, stability, and within-user null metrics, with explicit handling of constant vectors and inactive oracles; harmful actions are represented by their negative realized effect rather than a separate headline aggregate.
(3) We give executable algorithms for paired Monte Carlo attribution, signed action selection, the exact $B \leq 2$ oracle, user-level inference, and independent convergence selection.

(4) We report a two-dataset benchmark with ItemKNN as the primary quality-gated model and profile, full-catalogue, and sequential-scorer analyses as robustness boundaries. The primary results show that Shapley can improve relative to deletion while LIME and LOO retain higher absolute alignment and downstream decision quality in some conditions.

The paper makes no a priori claim that Shapley, LIME, LOO, or any other explainer is best. The framework is the contribution; method rankings are empirical and bounded by the declared offline simulator.

The paper is organized as follows. Section 2 surveys related work; Section 3 formalizes the temporal user-specific game; Section 4 defines the attribution methods, metrics, decision rule, and inference; Section 5 specifies the experimental protocol; Section 6 reports the results, including the hypothesis adjudication; Section 7 discusses implications and limitations. Extended confirmatory tables, replication summaries, and robustness matrices are in the supplementary material (Sections S1–S10), included with this submission.

2 Literature Review

2.1 Explainable recommendation

Recommendation explanations take several distinct forms—feature- and item-level rationales, relational paths, attention highlights, sentiment phrases, and natural-language text—and surveys of this literature stress that these forms serve different objectives, from user persuasion to model fidelity, transparency, and preference communication [77]. Early systems frequently produced explanations together with the recommendation itself: phrase-level sentiment models justify items through textual reviews [78], while latent-factor approaches expose item or feature contributions derived from the recommender's own decomposition [53, 55].

Post-hoc attribution transfers model-agnostic tools into the same domain, most prominently LIME [59] and SHAP [44]. Two recent developments sharpen the fidelity side of this picture. Barkan et al. [7] train the explanation mechanism itself against path-integration objectives across several recommender architectures and show that unrealistic attribution baselines can distort audits on sparse recommendation data, and Li et al. [41] develop counterfactual reasoning for path-based explanations that goes beyond attention scores. ActionShap addresses a complementary question: not how to generate an explanation, but whether a given explanation predicts the outcome of a declared feasible change.

2.2 Recommender models and offline evaluation

The offline recommendation literature supplies the model families and evaluation conventions on which the audit operates. Matrix factorization [38] and pairwise ranking objectives such as BPR [58] remain standard baselines, alongside learning-to-rank methods that formalize ranking quality directly [11, 34]. Neural architectures subsequently broadened the design space: neural collaborative filtering [27], recurrent session models [30], self-attentive sequential recommenders such as SASRec [36] and BERT4Rec [66], and graph models including NGCF [73] and LightGCN [26]. Graph recommenders are particularly relevant here because they separate a trained graph representation from what can be modified at inference time—the same distinction ActionShap draws between model parameters and inference-time profile weights. A parallel literature explains graph models themselves, through local surrogate subgraphs [45, 75], subgraph-level Shapley explorations [19, 76], and counterfactual graph explanations [42]; these methods generate explanations, whereas ActionShap audits any attribution—including graph-derived ones—against a declared intervention policy. We nevertheless adopt ItemKNN as the primary model because its dependence on the retained history is transparent under masking; the stochastic profile aggregator and quality-gated SASRec-style replication are robustness boundaries, not claims of broad architectural coverage.

The validity of offline evaluation is itself known to hinge on temporal leakage, negative sampling, exposure, and the candidate universe, and MovieLens [25] and the Netflix Prize [8] exemplify two widely used but structurally different feedback settings. Counterfactual evaluation adds the requirement of explicit assumptions about exposure and logged feedback [62]. A complementary line treats offline evaluation itself as a causal-inference problem: inverse-propensity and doubly-robust estimators correct the exposure bias of logged feedback [35, 61, 64], and bandit-style off-policy evaluation estimates the value of recommendation policies without live experiments [40]. ActionShap addresses a different bias—the mismatch between the perturbation an attribution is scored with and the intervention the system can execute—but shares this line's insistence that evaluation assumptions be stated and testable. ActionShap inherits the offline-evaluation cautions above: it fixes the temporal split, the candidate sets, and the intervention before evaluation, and it reports sampled-ranking and full-catalogue conditions separately.

2.3 Faithfulness, counterfactual evaluation, and recourse

Faithfulness diagnostics for attribution matured around deletion- and insertion-style tests. Ordered removal and restoration of input units quantify their influence on the model output [56], but the conclusions depend on the replacement baseline and on how perturbations are constructed [20], and aggressive perturbations can move inputs away from the data manifold [31]. Sanity checks provide a further diagnostic: Adebayo et al. [1] show that attribution maps can appear plausible while failing model- or label-randomization tests.

In recommender systems, a parallel literature generates counterfactual explanations. PRINCE searches provider-side profile changes that alter the recommendation list [21], ACCENT masks user factors to explain top-K outcomes [71], other formulations define counterfactuals directly over user evidence or item aspects [6, 70], and learned explanation models extend the idea across recommendation instances [24]. Recent generators push toward minimal or plausible perturbations, using guided denoising diffusion for minimal-perturbation counterfactuals [48] or plausibility-constrained explanations validated with a user study [13]. Algorithmic recourse asks a broader question—which feasible feature changes would flip a consequential decision [37, 72]—and ActionShap deliberately does not claim causal recourse or user authority over the audited profile weights.

A smaller but directly relevant literature evaluates counterfactual explanations rather than generating them. Early benchmarks tested whether explanations predict recommendation changes [4, 74]; later studies documented inconsistencies in top-k evaluation protocols [47] and built broader reproducibility benchmarks [51]. Two positions frame our design: automated counterfactual metrics do not exhaust explanation quality, so human-centric assessment remains necessary [18], while LLM-powered benchmarks such as ALERT make explanation assessment increasingly scalable [39]. Path-integration attribution supplies the continuous-path reference against which our discrete bounded-weight semantics can be compared [69]. ActionShap differs in evaluating an attribution under one fixed, declared intervention policy and in making the policy, cost, authority, and offline limitations explicit.

2.4 Evaluation protocols for recommender explanations

A distinct strand treats explanation evaluation as a protocol rather than a generation task; systematic reviews catalog the available quantitative evaluation methods [50], and open toolkits implement large families of them [28]. Rank-oriented protocols measure attribution fidelity for ordered outputs, through rank-aware Shapley formulations [14] or rank-based fidelity scores [57]; sequence-oriented protocols adapt event deletion to sequential outputs [9] or audit session-model attributions [10]; further benchmarks target contrastive completeness [60] or automatic natural-language assessment [39].

Two empirical findings from this strand shape ActionShap directly. First, Mohammadi et al. [46] show that recommender performance itself changes the ranking of explainers, so explanation evaluation should be gated on recommendation quality; ActionShap therefore audits model quality against popularity before any attribution result is interpreted (Section 6.6 shows why the gate matters). Second, fidelity measured under one perturbation semantics need not transfer to another [5, 7], which motivates separating deletion alignment from bounded-intervention alignment rather than treating a single deletion score as explanation validity. Against these protocols, ActionShap differs in three respects: the perturbation is a declared system-executable weight policy rather than a feature- or event-deletion convention; joint actions are scored against an exact common budget oracle; and inference is performed over distinct users with within-user null calibration.

2.5 Interpretable machine learning and cooperative attribution

Interpretability is not a single property: taxonomies distinguish faithfulness, stability, simulatability, and usefulness [3, 23], and empirical benchmarks show that these properties can disagree in practice [17, 63]—a tension echoed across methods in broader syntheses [49]. This disagreement is one motivation for reporting several clearly separated metrics rather than a single faithfulness score.

Shapley attribution grounds feature attribution in cooperative-game values [65]; SHAP adapts the construction to post-hoc model explanation [43, 44]; and sampling- and regression-based estimators make it applicable when exact enumeration is infeasible [12, 15, 54]. Several distinct Shapley-style values exist for model explanation depending on the value function and intervention [68]; ActionShap declares its value function explicitly (the target-margin utility over weight vectors) and uses Monte Carlo permutation Shapley for it, with additive global importance measures as a related family [16]. Gradient-based attributions admit the same weight-space interface and are represented here through the finite-difference and integrated-gradient baselines evaluated in Supplementary Section S6 [2]; causal variants of Shapley values instead exploit causal structure [29], which ActionShap does not assume or estimate. The cost of exact computation has itself become a research object: amortized estimators learn to predict Shapley values in real time [33], and data-valuation work treats training points as players whose Shapley values quantify their contribution [22]—the same player abstraction ActionShap applies to interaction records. ActionShap treats Shapley as one interchangeable explainer among several: the Shapley axioms do not imply causal validity, intervention feasibility, or decision superiority, and the protocol measures each of those properties separately.

Table 1. Positioning by evaluation target.

Approach	Explainer audit	Declared executable policy	Joint regret	Abstention
Deletion/insertion	Yes	No	No	No
Counterfactual expl.	Usually	Method-spec.	Sometimes	Method-spec.
Algorithmic recourse	No	Yes	Optim. target	Often
ActionShap	Yes	Yes	Yes	Yes

3 Background and Problem Formulation

3.1 Temporal user-specific game

For user u, let H_u^{train} contain all interactions before the held-out validation and test events. The player set is the most recent training window,

$$P_u = \{p_{u1}, \dots, p_{un_u}\}, \qquad n_u \le n_{\max} = 20, \tag{1}$$

and define the older training context $B_u = H_u^{\text{train}} \setminus P_u$. Interactions are ordered by (timestamp, original_record_index); the final event is the test target and the preceding event is validation. Each player is a distinct interaction record in this ordering, so repeated interactions with the same item remain distinct. Users with fewer than four interactions overall are ineligible. After reserving validation and test, an eligible user therefore has at least two training interactions; the observed minimum is three players, represented by the 3–5 stratum in Table 10. The only truncation rule retains the most recent $\min(|H_u^{\text{train}}|, n_{\max})$ records, and validation and test events are never players. Models are fitted on the complete training histories H_u^{train}, but inference-time scoring excludes the older context B_u: the profile score $f_u(i; S)$ depends only on S. Consequently, $n_{\max}$ defines both the operating profile and the player game; the $n_{\max} \in \{50, 100\}$ runs vary that boundary rather than holding the recommender input fixed.

Every user has a fixed candidate set E_u containing the held-out target and sampled negatives. Negatives exclude the complete pre-test history, including validation. The user seed, candidate seed, tie seed, model seed, and attribution seed are independent. Full-catalogue robustness uses the target plus all unseen items. Target coverage is one by construction and is not called retrieval recall.

3.2 History-conditioned recommenders

The primary ItemKNN score is a weighted mean of frozen item–item similarities. A weight vector $w \in [0,1]^{P_u}$ assigns an inference-time weight to each retained interaction; a weighted coalition is a pair (S, w) with $S \subseteq P_u$, and we define $W_u(S, w) = \sum_{h \in S} w_h$:

$$f_u(i \mid S, w) = \begin{cases} \dfrac{\sum_{h \in S} w_h s(i, h)}{W_u(S, w)}, & W_u(S, w) > 0, \\ 0, & W_u(S, w) = 0. \end{cases} \tag{2}$$

A profile aggregator is used for robustness,

$$p_u(S, w) = \begin{cases} \dfrac{\sum_{h \in S} w_h q_h}{W_u(S, w)}, & W_u(S, w) > 0, \\ \mathbf{0}, & W_u(S, w) = 0, \end{cases} \qquad f_u(i \mid S, w) = q_i^\top p_u(S, w). \tag{3}$$

Thus the empty coalition and any nonempty all-zero-weight coalition use the same zero-profile convention. The primary full profile uses $w_h = 1$ for every retained interaction. All utilities below are functions of the weighted coalition (S, w); for $z \in \{v_u^{\text{attr}}, q_u\}$ we write $z_u(S, w)$ and abbreviate $z_u(S) := z_u(S, \mathbf{1})$, $f_u^S(i) := f_u(i \mid S, \mathbf{1})$, and $s_i^{S,w} := f_u(i \mid S, w)$ with $s_i^S := s_i^{S,\mathbf{1}}$. The robustness model's item embeddings are trained once with the leave-one-out BPR-style objective

$$\mathcal{L}_{\text{BPR}} = -\sum_{(u,i,j)} \log \sigma\left(r_{u,-i}^\top (q_i - q_j)\right) + \frac{\lambda}{2}\left(\|q_i\|_2^2 + \|q_j\|_2^2 + \sum_{h \in H_u^{\text{train}} \setminus \{i\}} \|q_h\|_2^2\right) \quad \text{(applied per gradient step)}, \tag{4}$$

where the triple (u, i, j) is sampled with u uniform over eligible users, i uniform over u's distinct training items, and j uniform outside that history; $r_{u,-i} \in \mathbb{R}^d$ is the mean of the other complete-training-history embeddings (renamed to avoid overloading the player symbol p), $\lambda = 10^{-4}$, the embedding dimension is 64, and ten epochs with one sampled positive per user per epoch are used. Item embeddings are initialised from $\mathcal{N}(0, 0.05^2)$ and updated by plain stochastic gradient descent with learning rate 0.03; each update batch is one triple (sampled positive, uniform negative, leave-one-out context) per user per epoch, and the L_2 penalty is applied to every embedding participating in the triple (positive, negative, and context vectors), with margins clipped to ±35 for numerical stability. Users with fewer than two distinct training items contribute no triple and therefore do not update the fit; eligibility for the evaluation cohort is governed by the four-interaction floor of Section 3.1. Only item embeddings are retained: the deployed user representation is recomputed from the history supplied at inference time. Static user embeddings are excluded from attribution because post-training history masking cannot change their scores.

3.3 Utilities and interventions

The primary characteristic function is the target-margin utility

$$v_u^{\text{attr}}(S, w) = \sigma\left(s_y^{S,w} - \frac{1}{L} \sum_{i \in \text{TopL}_{-y}(S,w)} s_i^{S,w}\right), \tag{5}$$

where y is the temporal target, $L = 10$, and $\sigma(t) = [1+\exp(-t)]^{-1}$ has unit temperature. The competitor set $\text{TopL}_{-y}(S, w)$ is recomputed from the current weighted-coalition scores for every (S, w); it contains the ten highest-scoring candidates in $E_u \setminus \{y\}$. Because $\text{TopL}_{-y}(S, w)$ changes discontinuously, v_u^{attr} is piecewise smooth and generally non-monotone; this does not affect the Shapley construction but matters for interpretation and for gradient-based baselines. Every retained positive interaction starts with inference-time weight $w_h = 1$. No temporal decay or rating-dependent weight is used. An action changes only these inference-time weights and never retrains the model. For the fixed candidate set E_u, let τ_i be the global tie priority and define the one-indexed target rank under a weighted coalition,

$$r_u(y \mid S, w) = 1 + \sum_{i \in E_u \setminus \{y\}} \mathbf{1}[s_i^{S,w} > s_y^{S,w}] + \sum_{i \in E_u \setminus \{y\}} \mathbf{1}[s_i^{S,w} = s_y^{S,w} \ \wedge \ \tau_i < \tau_y]. \tag{6}$$

With one relevant target, $IDCG@10 = 1$ and the operational ranking outcome is

$$q_u(S, w) = \text{NDCG}@10(f_u(\cdot \mid S, w), y) = \mathbf{1}[r_u(y \mid S, w) \le 10] \, \frac{1}{\log_2(r_u(y \mid S, w) + 1)}. \tag{7}$$

Thus HitRate@10 is the one-target hit indicator, not catalogue retrieval recall. All effects and regrets are stored separately for target margin and NDCG. The empty coalition has target-margin utility 0.5 and deterministic catalogue ties.

For an action $A \subseteq P_u$, let $w^{A,\rho} \in [0, 1]^{P_u}$ denote the weight vector

$$w_p^{A,\rho} = \begin{cases} \rho, & p \in A, \\ 1, & p \notin A, \end{cases} \tag{8}$$

and define the signed intervention effect for either utility $z \in \{v_u^{\text{attr}}, q_u\}$ by

$$\Delta_u^z(A; \rho) = z_u(P_u, w^{A,\rho}) - z_u(P_u, \mathbf{1}). \tag{9}$$

Deletion and bounded singleton effects are therefore

$$\Delta_{u,p}^{z,\text{del}} = \Delta_u^z(\{p\}; 0), \qquad \Delta_{u,p}^{z,\text{bnd}} = \Delta_u^z(\{p\}; 0.5). \tag{10}$$

At $\rho = 0$ the intervention assigns weight zero to p, so deletion is evaluated with the explicit zero-profile convention of Eqs. (2) and (3) on the identical coalition-evaluation path as leave-one-out. The primary feasible intervention is bounded downweighting,

$$\mathcal{I}_\rho(p): \ w_p \leftarrow \rho w_p, \qquad \rho = 0.5, \tag{11}$$

with $\rho = 0.25$ as a declared sensitivity. $\mathcal{I}_\rho(p)$ denotes a simulator-side weight intervention, not a causal do-operation. No model retraining occurs after an intervention.

The action space is

$$\mathcal{A}_{u,B} = \{A \subseteq P_u : |A| \leq B\}, \tag{12}$$

including no action. The primary scientific budget is $B = 2$. A signed attribution predicts downweighting benefit as $-\phi$; only positive predicted benefit is selected. LOO is an algebraic deletion oracle at $B = 1$, not a headline competitor.

4 Methodology

4.1 Protocol architecture

Figure 1 summarizes the evaluation pipeline used throughout the paper. The architecture is placed with the methodology because it defines the sequence of player construction, scoring, attribution, intervention, and outcome measurement before the experimental results are introduced.

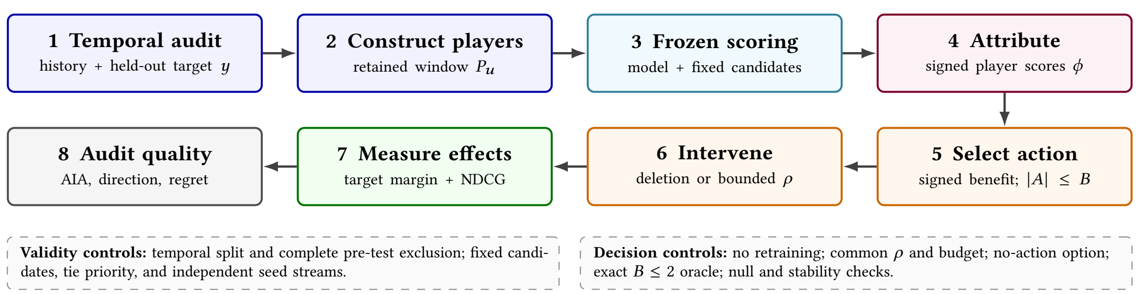

Fig. 1. ActionShap evaluation architecture for a retrospective, target-conditioned audit. The held-out temporal target y defines the target-margin utility; the main path keeps player construction, frozen scoring, attribution, action selection, intervention, and outcome evaluation distinct. Control bands identify quantities fixed across explainers; stability diagnostics appear in the diagnostics tables in Supplementary Section S8.

4.2 Attribution methods

For a sampled base permutation π_m, let S_{p,π_m} be the predecessor coalition of player p. We use M_{pair} base permutations and also evaluate each reverse order, giving $T = 2M_{\text{pair}}$ total prefix walks. The estimator is

$$\widehat{\phi}_{u,p} = \frac{1}{T} \sum_{t=1}^{T} \left[v_u^{\text{attr}}(S_{p,\pi_t} \cup \{p\}) - v_u^{\text{attr}}(S_{p,\pi_t}) \right]. \tag{13}$$

The exact value for reference is defined on the game $G_u = (P_u, v_u^{\text{attr}})$ with $v_u^{\text{attr}}(\varnothing) = 0.5$, so efficiency sums to $v_u^{\text{attr}}(P_u) - 0.5$; it is

$$\phi_{u,p}^{\text{exact}} = \sum_{S \subseteq P_u \setminus \{p\}} \frac{|S|!\,(n_u - |S| - 1)!}{n_u!} \left[v_u^{\text{attr}}(S \cup \{p\}) - v_u^{\text{attr}}(S) \right]. \tag{14}$$

As a cooperative-game value, this exact expression satisfies efficiency, symmetry, dummy, and additivity; these classical properties are standard and are not re-derived. The reverse-paired estimator above is unbiased for this factorial-weighted value; it is used because n_u can be too large for enumerating all 2^{n_u} coalitions. The reverse orders are reverse-paired samples used as a variance-reduction heuristic (reported as reverse pairing; a variance comparison against unpaired sampling is provided in the release), and the coalition-value cache is shared between a permutation and its reverse. In the executable configuration, `permutations` denotes $M_{\text{pair}} = 500$, hence $T = 1000$ evaluated orders. The comparison includes locally weighted binary-mask LIME, LOO, greedy sequential deletion, and random attribution. No method observes measured intervention effects, oracle actions, or aggregate test outcomes.

For LIME, a binary mask $m_b \in \{0,1\}^{n_u}$ defines coalition $S_b = \{p : m_{b,p} = 1\}$ and response $y_b = v_u^{\text{attr}}(S_b)$. With normalized Hamming distance $d_b = n_u^{-1} \sum_p (1 - m_{b,p})$, kernel width $\kappa = .25$, and ridge coefficient $\lambda_{\text{LIME}} = 1$, the fitted local surrogate is

$$(\hat{\beta}_0, \hat{\boldsymbol{\beta}}) = \arg\min_{\beta_0, \boldsymbol{\beta}} \sum_{b=1}^{512} \exp\left(-d_b^2/\kappa^2\right) \left(y_b - \beta_0 - m_b^\top \boldsymbol{\beta}\right)^2 + \lambda_{\text{LIME}} \|\boldsymbol{\beta}\|_2^2. \tag{15}$$

The intercept is unpenalized and $\hat{\boldsymbol{\beta}}$ is the LIME attribution. The penalised least-squares problem is solved directly by closed-form ridge regression (deterministic; no iterative initialisation). The mask design contains the full, empty, and every leave-one-out mask; the remaining rows are independent Bernoulli(0.5) masks (each history interaction retained with probability 1/2), drawn from the attribution seed stream, which is independent of the model, candidate, tie, and user seeds. Masks are drawn with replacement: for short profiles the number of distinct coalitions is far below the design size ($2^5 = 32$ unique coalitions against 512 rows at Amazon's median $n_u = 5$), so repeated masks occur and the closed-form ridge fit implicitly weights each coalition by its multiplicity. We keep this identical seeded design at every history length for comparability and record it in the frozen configuration; enumeration-based and without-replacement mask designs are listed as a validation extension (Section 7.1).

For completeness, the greedy and random controls are defined operationally. Greedy sequential deletion starts at $S_0 = P_u$. At step t it evaluates $v_u^{\text{attr}}(S_{t-1} \setminus \{p\}) - v_u^{\text{attr}}(S_{t-1})$ for every remaining player, selects the largest benefit, breaks ties by the lowest player index, records the signed value under the convention that $-\phi$ predicts downweight benefit, and recomputes the next step from the reduced coalition. It repeats until every player is ordered; it does not inspect bounded effects or the oracle, and the later signed action rule may abstain. Gradient-based attributions are available wherever the scoring path is differentiable with respect to the retained weights: finite-difference and integrated-gradient attributions along the weight path are evaluated in the replication subsection (Supplementary Section S6) for both models, where they act as path-matched, intervention-aware baselines. Random attribution draws independent scores $r_{u,p} \sim \mathcal{N}(0,1)$ for every user and experiment seed using NumPy's seeded generator. The random vector is passed through the same signed action-selection rule as every explainer, including the same budget and abstention option. Its seed is independently offset from the model/attribution seed by $1{,}000{,}000 + u$, so it is redrawn per user and seed.

4.3 Metrics

Let $\rho_S(\cdot,\cdot)$ denote Spearman rank correlation. Magnitude AIA is

$$\text{AIA}_u^z(g) = \rho_S(|\phi_{u,p}^g|, |\Delta_{u,p}^{z,\text{bnd}}|), \qquad z \in \{v_u^{\text{attr}}, q_u\}. \tag{16}$$

Here and throughout, the superscript $g \in \{\text{Shapley}, \text{LIME}, \text{LOO}, \text{Greedy}, \text{Random}\}$ indexes the attribution method. Deletion AIA substitutes $\Delta^{z,\text{del}}_{u,p}$; bounded AIA uses $\Delta^{z,\text{bnd}}_{u,p}$, and their difference is the bounded-minus-deletion magnitude-alignment difference (short: the gap). We use Spearman rank correlation rather than Pearson because the mapping from effects to utility is nonlinear (sigmoid margin, discrete NDCG) and heavy-tailed per-user effects would let a few extreme players dominate a linear correlation; Spearman also makes the metric invariant to monotone re-expressions of the utility. Kendall's τ is a rank-equivalent alternative and yields qualitatively identical orderings on these data; we retain Spearman for comparability with the deletion-evaluation literature. Signed alignment is $\rho_S(-\phi^g_{u,p}, \Delta^z_{u,p})$. For informative non-zero effects $I^z_u = \{p : \Delta^z_{u,p} \neq 0\}$, direction accuracy is

$$\text{DirAcc}^z_u(g) = |I^z_u|^{-1} \sum_{p \in I^z_u} \mathbf{1}\left[\text{sign}(-\phi^g_{u,p}) = \text{sign}(\Delta^z_{u,p})\right]. \tag{17}$$

We adopt $\text{sign}(0) = 0$; zero effects are excluded from I^z_u, so the value of $\text{sign}(0)$ never contributes. When $I^z_u = \varnothing$, direction accuracy is missing. Let $k_u = \min(k, n_u)$ and let T^g_{u,k_u} be the k_u players with largest predicted benefit $-\phi^g$; top-k intervention precision is reported for $k \in \{1, 3, 5\}$ in the release:

$$\text{Prec@}k^z_u(g) = k_u^{-1} \sum_{p \in T^g_{u,k_u}} \mathbf{1}\left[\Delta^z_{u,p} > 0 \ \wedge \ \text{sign}(-\phi^g_{u,p}) = \text{sign}(\Delta^z_{u,p})\right]. \tag{18}$$

The indicator's sign condition is redundant given $\Delta > 0$ and is kept for auditability; Prec@k is a fixed-k diagnostic over the k largest predicted benefits and is deliberately distinct from the abstention rule. If $n_u = 0$ the quantity is missing. For a selected joint action $\widehat{A}_{u,g}$, success is $\mathbf{1}[\Delta^z_u(\widehat{A}_{u,g}) > 0]$; a negative realized effect identifies a harmful action, and abstention is $\mathbf{1}[\widehat{A}_{u,g} = \varnothing]$. Rank stability over R_{seed} seed runs is

$$\text{Stab}_u(g) = \frac{1}{|V_u|} \sum_{(r,s) \in V_u} \rho_S(\boldsymbol{\phi}^{g,r}_u, \boldsymbol{\phi}^{g,s}_u), \qquad V_u = \{(r,s) : 1 \le r < s \le R_{\text{seed}}, \text{ both vectors finite and nonconstant}\}, \tag{19}$$

computed only over finite nonconstant pairs; users with fewer than two such pairs are missing for this diagnostic.

For each user and seed, $R_{\text{null}} = 1{,}000$ within-seed permutations shuffle effect magnitudes across players. We first compute the observed AIA and its R_{null} null draws within each seed; then we average the observed value and each matched null draw across the five seeds within user. A user-level estimate requires at least three of five valid seed-level values; users with fewer are excluded and reported as missing. In the primary runs every retained user has all five seeds valid, and the seven Amazon users missing under every seed are excluded ($n = 993$). Bootstrap and paired inference operate on these distinct-user summaries. The null test is a one-sided upper-tail test of alignment beyond chance, defined at the dataset level to match Table 6. For each valid user u, let $\bar{A}^{\text{obs}}_u$ be the magnitude AIA averaged over the five seeds and $\bar{A}^{(r)}_u$ the matched seed-averaged null draw r (effect magnitudes shuffled across players within each user and seed). With U valid users,

$$p_{\text{null}} = \frac{1 + \sum_{r=1}^{R_{\text{null}}} \mathbf{1}\left[U^{-1} \sum_u \bar{A}^{(r)}_u \ \ge \ U^{-1} \sum_u \bar{A}^{\text{obs}}_u\right]}{R_{\text{null}} + 1}, \tag{20}$$

i.e. the dataset-level observed alignment is compared against the dataset-level null distribution of seed-averaged draws. The ten dataset–method null tests in Table 6 are unadjusted descriptive calibration controls (labelled "Unadj. upper-tail null p"), not a confirmatory multiplicity family; confirmatory method comparisons use the Holm-adjusted paired tests of the statistical-validation tables in Supplementary Section S8 and Supplementary Tables S3–S5. We report null means, null 95th percentiles, plus-one permutation p-values, and valid-user counts. Constant vectors are missing, not imputed.

4.4 Decision quality and regret

For a method action, define the additive predicted-benefit score

$$\widehat{B}_u^g(A) = -\sum_{p\in A} \widehat{\phi}_{u,p}^g. \tag{21}$$

This is an attribution-ranked additive decision heuristic, not a calibrated estimate of the realized joint effect. Realized pair effects may include interactions $I_{pq} = \Delta(\{p,q\}) - \Delta(\{p\}) - \Delta(\{q\})$ [32, 67]; Supplementary Section S6 quantifies the resulting interaction error and evaluates an interaction-aware alternative that re-scores candidate pairs before selection, which changes the chosen action precisely when interactions are large (the SASRec runs). The additive rule is retained as the primary decision rule because it is the canonical budget-B policy implied by any additive attribution and because it is the rule a practitioner would apply from an attribution vector alone; the eligibility threshold $\varepsilon = 10^{-12}$ is a numerical zero tolerance that separates exactly-neutral candidates from sign-informative ones, and the results are insensitive to its value over $[10^{-15}, 10^{-6}]$ because signed effects are either identically zero or orders of magnitude larger. The method considers an action eligible when $\widehat{B}_u^g(A) > 10^{-12}$; otherwise it abstains. Because this predicted benefit is additive, the implementation obtains the maximizer by sorting strictly positive individual benefits and taking at most B of them; equal scores use the deterministic tie rule below. For utility z, the exact budget-two oracle is

$$A_u^{*,z} = \arg\max_{A\in\mathcal{A}_{u,2}} \Delta_u^z(A), \tag{22}$$

and the method action is $\widehat{A}_{u,g}$. Because the no-action set belongs to $\mathcal{A}_{u,2}$, the oracle effect satisfies $\Delta_u^z(A_u^{*,z}) \geq 0$ by construction, so the normalized-regret denominator in Eq. (25) is only ever guarded against numerical zero, never negative. Regret is

$$\mathrm{Regret}_u^z(g) = \Delta_u^z(A_u^{*,z}) - \Delta_u^z(\widehat{A}_{u,g}). \tag{23}$$

For a general budget B, the utility-specific feasible oracle is

$$A_u^{*,z}(B) = \arg\max_{A\subseteq P_u,\,|A|\leq B} \Delta_u^z(A). \tag{24}$$

At $B = 2$ the released primary results exhaustively enumerate no action, all singletons, and all pairs. The archived $B = 3$ sensitivity was produced with the declared greedy lower-bound search rather than exhaustive triples; it is retained only for joint-effect, success, and abstention sensitivity. No normalized-regret claim is made for $B = 3$. Exhaustive triples would require at most $1 + 20 + \binom{20}{2} + \binom{20}{3} = 1351$ actions per user and are recommended for a future regenerated $B = 3$ release.

For utility z, normalized regret is

$$\mathrm{NRegret}_u^z(g) = \frac{\Delta_u^z(A_u^{*,z}) - \Delta_u^z(\widehat{A}_{u,g})}{\Delta_u^z(A_u^{*,z})}, \qquad \Delta_u^z(A_u^{*,z}) > \varepsilon, \quad \varepsilon = 10^{-12}. \tag{25}$$

It is averaged only over active-oracle users. Zero means an oracle-equivalent action; one means that the oracle improves the utility but the selected action achieves no improvement; values above one mean that the selected action is harmful. Values below zero are rejected by validation for the exact shared action space (up to floating-point tolerance); they would indicate an oracle or action-space inconsistency. The quantity is not clipped to one.

Equal action scores use a deterministic common rule: maximize predicted benefit first; among ties choose the smaller action cardinality; among equal-cardinality ties choose the lexicographically lower sorted player-index tuple. An

Algorithm 1 Paired Monte Carlo attribution

Require: $u, P_u, v, M_{\text{pair}}$, attribution seed
Ensure: $\widehat{\boldsymbol{\phi}}_u$

1: Initialise $\hat{\phi}_{u,p} \leftarrow 0$ for all $p \in P_u$
2: **for** $m = 1$ to M_{pair} **do**
3: Draw base order $\pi_m \sim \text{Uniform}(\Pi(P_u))$; evaluate π_m and its reverse as cached prefix walks
4: **for** each consecutive pair $(S, S \cup \{p\})$ **do**
5: $\hat{\phi}_{u,p} \leftarrow \hat{\phi}_{u,p} + v(S \cup \{p\}) - v(S)$
6: **end for**
7: **end for**
8: Divide by $T = 2M_{\text{pair}}$ and record finite/constant-vector flags

Algorithm 2 Leakage-free signed action selection and exact budget-two oracle

Require: attribution $\hat{\boldsymbol{\phi}}_u$, player set P_u, budget $B = 2$, strength ρ, candidate set E_u, deterministic tie rule, utilities $z \in \{v_u^{\text{attr}}, q_u\}$; information set $\mathcal{I}_{\text{select}} = \{\hat{\boldsymbol{\phi}}_u, P_u, \rho, B, \tau\}$ with $\Delta_u^z(A) \notin \mathcal{I}_{\text{select}}$
Ensure: frozen method action, oracle actions, realized effects

1: **Stage 1 (selection, outcome-blind):** $\widehat{B}_u^g(A) \leftarrow -\sum_{p \in A} \hat{\phi}_{u,p}$ for $A \in \mathcal{A}_{u,B}$
2: $\widehat{A}_{u,g} \leftarrow$ argmax of eligible $\widehat{B}_u^g$ (deterministic tie rule); $\widehat{A}_{u,g} \leftarrow \varnothing$ if best score $\leq 10^{-12}$
(additive unit-cost scoring makes sorting positive individual benefits equivalent to enumerating $\mathcal{A}_{u,B}$; the additive score is a decision heuristic, not a consequence of Shapley theory)
3: **FREEZE** $\widehat{A}_{u,g}$
4: **Stage 2 (outcome evaluation, after freezing):** for $A \in \{\varnothing\} \cup$ singletons $\cup$ unordered pairs: construct $w^{A,\rho}$ and compute $\Delta_u^z(A)$ for every z
5: $A_u^{*,z} \leftarrow \arg\max_A \Delta_u^z(A)$ with cardinality-then-lexicographic tie rule
6: **return** $\widehat{A}_{u,g}$, $\{A_u^{*,z}\}_z$, $\{\Delta_u^z(A)\}_{A,z}$

Algorithm 3 Distinct-user inference and AIA null

Require: five seed records per user and $R_{\text{null}} = 1000$ null draws per seed
Ensure: User-level estimates and corrected comparisons

1: **for** each distinct user u and seed s **do**
2: Compute seed-level AIA only for finite, nonconstant vectors
3: **for** $r = 1$ to R_{null} **do**
4: Shuffle effect magnitudes within (u, s); recompute the null AIA
5: **end for**
6: **end for**
7: **for** each distinct user u **do**
8: Require at least three of five valid seed-level values, else mark missing
9: Average the observed AIA and each matched null draw across valid seeds
10: **end for**
11: Bootstrap distinct users ($R_{\text{boot}} = 10{,}000$); for each declared paired comparison run $R_{\text{perm}} = 10{,}000$ sign permutations, plus-one two-sided p, Holm correction within the explicitly named dataset–model–condition–metric family

abstention action wins when the best predicted benefit is at most 10^{-12}. The exact oracle uses the same cardinality and lexicographic rule after maximizing realized utility.

Algorithm 4 Independent convergence selection

Require: $M_{\text{pair}} \in \{25, 50, 100, 250, 500, 1000\}$ and an independent $M_{\text{pair}} = 1000$ reference
Ensure: Selected M_{pair} or explicit unconverged status
1: **for** each candidate M_{pair} **do**
2: Evaluate $T = 2M_{\text{pair}}$ forward/reverse orders and compare attribution ranks and signed-$B = 2$ action Jaccard with the independent reference
3: Record rank-valid fraction and threshold coverage
4: **end for**
5: Select smallest M_{pair} whose population means satisfy rank $\geq .95$, Jaccard $\geq .80$, rank validity $\geq .95$ (a population-average diagnostic; per-user coverage is reported separately, so conditions are labelled mean-converged rather than user-universally converged)
6: **if** no candidate qualifies **then**
7: Retain $M_{\text{pair}} = 1000$ as an unconverged stress test
8: **end if**

Table 2. Primary-cohort profile-window and intervention-opportunity audit. "At cap" is the fraction with $n_u = 20$; an active NDCG oracle has a positive exact budget-two effect.

Dataset	Users	Mean n_u	Median n_u	At cap	Active oracle
MovieLens-1M	1000	18.81	20	83.1%	339
Amazon Digital Music	1000	6.66	5	5.9%	196

5 Experimental Protocol

5.1 Datasets and temporal split

The selected evaluation cohort comprises 1,000 users per dataset drawn with user seed 2718 from the eligibility pool of test users with at least four interactions overall: 6,028 eligible users on MovieLens-1M and 7,621 on Amazon Digital Music. The cohort is therefore a seeded subsample of a fully enumerated pool rather than an identifier-ordered prefix. After validation and test are reserved, the minimum possible player count is two; the observed minimum is three. Retained history lengths differ sharply between domains (MovieLens median/IQR 20/20 events at the $n_{\max}$ cap, Amazon 5/3–8, reflecting its sparsity); per-step filtering counts are recorded in the release dataset audit. The audit table makes the cohort asymmetry explicit: most MovieLens users sit at the 20-player boundary, whereas the median Amazon user has five players and only 5.9% reach the cap. Cross-dataset contrasts are therefore interpreted descriptively, and Section 6.6 reports a history-stratified analysis rather than attributing the larger Amazon difference to dataset identity alone. We use MovieLens-1M [25] and Amazon Digital Music reconstructed from the Amazon Reviews 2018 release [52]. Positive feedback is rating ≥ 4; Amazon duplicate user–item records retain the latest deterministic event and iterative 5-core filtering is reapplied after thresholding. MovieLens retains 6,035 users, 3,533 items, and 575,276 positive interactions. Amazon retains 10,623 users, 8,582 items, and 104,431 interactions. Source and processed hashes are recorded in the final manifest.

5.2 Models, candidates, and gates

The matrix includes ItemKNN and profile aggregation, five seeds, primary 1,000-user sampled ranking, 250-user full-catalogue subsets, and declared history, candidate, intervention-strength, budget, and utility sensitivities. ItemKNN uses cosine item–item similarities from binary training histories, removes self-similarity, retains the 200 largest non-zero neighbours per item, and applies no shrinkage or minimum-support threshold beyond the observed co-occurrence counts; all retained similarities are non-negative. The profile robustness model uses 64-dimensional item embeddings, ten leave-one-out BPR-style epochs, one sampled positive per user per epoch, uniform negatives outside the complete training history, learning rate 0.03, and L_2 coefficient 10^{-4}. Both models recompute the profile from the retained player window at inference time and are not retrained under an intervention.

The primary candidate set contains the target and 199 uniformly sampled unseen negatives. Candidate seed 1729, user seed 2718, tie seed 31415, and model/attribution seeds 42–46 are independent. The masking gate is defined as follows: for 200 gate users (seeded subsample) on a fixed 200-item sampled candidate set, exactly one uniformly chosen player is deleted per user; the gate passes for a model seed if (i) the top-10 order changes for at least 50% of gate users, (ii) the mean absolute NDCG@10 change is at least 10^{-3}, and (iii) a frozen-score static control is exactly invariant to the mask argument (verifying that only the retained-history path carries information). ItemKNN passes all blocking masking gates on every seed. The profile model fails the NDCG masking threshold for one MovieLens seed (mean absolute change 0.000682 versus 0.001) and is retained as a nonblocking robustness boundary.

LIME uses 512 binary masks including the full, empty, and all leave-one-out masks, a normalized Hamming-distance kernel with width 0.25, and ridge coefficient 1.0. The target-margin competitor count is $L = 10$ and the sigmoid temperature is one. The primary intervention is $\rho = 0.5$; $\rho = 0.25$ is sensitivity-only. The exact budget-two action space includes no action, all singletons, and all pairs.

5.3 Statistical analysis

For each seed we compute user-level quantities first; repeated seed values are then averaged within distinct user before bootstrap resampling. A user summary requires at least three of five valid seed-level values; users with fewer valid seeds are excluded and counted as missing (in the primary runs all retained users have five valid seeds, and seven Amazon users are missing under every seed). User bootstrap intervals are two-sided percentile intervals from 10,000 user resamples. For a paired metric, $d_u = x_u - y_u$ and $T_{\text{obs}} = n^{-1} \sum_{u=1}^{n} d_u$. Paired plus-one sign permutations use 10,000 two-sided sign flips $T_r^* = n^{-1} \sum_u \epsilon_{u,r} d_u$, with $\epsilon_{u,r} \in \{-1, +1\}$ and $p = (1 + \#\{|T_r^*| \geq |T_{\text{obs}}|\})/(R + 1)$. Holm correction is applied separately within each dataset–model–condition–metric family across the declared method-pair comparisons. Paired Cohen's d_z is $\bar{d}/s_d$, where d_u is the user-level paired difference and s_d its sample standard deviation. The paired comparison uses the intersection of users with finite values for both methods; missing correlations are retained as missing. Active-oracle counts define the denominator for normalized regret. No coalition evaluation is an independent statistical observation. Three further checks frame the confirmatory family. First, a Friedman omnibus test over the five methods' user-level bounded AIA rejects equal distributions on both datasets ($\chi^2 = 3767.2$ on MovieLens, 2482.0 on Amazon; both $p < 10^{-300}$ at $n \geq 993$), justifying the subsequent pairwise comparisons. Second, paired Cohen's d_z is reported with every Holm-adjusted comparison in the release; for the headline alignment contrasts it is large ($d_z = -1.010$ Shapley–LIME and -1.254 Shapley–LOO on MovieLens; -0.726 and -0.771 on Amazon), while for realized-NDCG contrasts it is small ($|d_z| \leq 0.085$). Third, a post-hoc minimum-detectable-effect analysis at 80% power shows the 1,000-user cohort resolves bounded-AIA differences of ±0.014 (MovieLens) and ±0.051 (Amazon), but resolves NDCG-effect differences only to ±0.002–0.004; the observed decision-quality differences (≤ 0.003) therefore fall at or below the detectable threshold, so non-detection of decision-quality differences is a power statement, not an equivalence claim (the TOST equivalence result for Amazon Shapley–LIME uses an independently declared margin).

5.4 Reproducibility

The result archive contains 80 recommendation runs and five convergence studies. All manuscript assets are generated from schema-v2 raw results using repository-relative paths and content hashes. The primary ItemKNN target-margin analysis uses $M_{\text{pair}} = 500$ base permutations and $T = 1000$ evaluated orders; although the independent convergence study selects lower diagnostic values for some conditions, the conservative final floor is retained. The archived runs record macOS-26.5.1-arm64, Python 3.12.13, NumPy 2.4.6, SciPy 1.17.1, pandas 2.3.3, scikit-learn 1.9.0, matplotlib 3.11.1, and PyYAML 6.0.3. Processor model, RAM, and peak RSS were not serialized and are not claimed.

6 Results

All reported results passed the frozen schema-v2 validation checks; pilot runs are excluded from the canonical analysis.

6.1 Recommendation quality and protocol audit

Table 3 shows that primary ItemKNN exceeds the popularity reference on both datasets. MovieLens ItemKNN reaches NDCG@10 = 0.284 and HitRate@10 = 0.504, compared with 0.158 and 0.320 for popularity. Amazon ItemKNN reaches 0.181 and 0.285, compared with 0.090 and 0.155. The profile model is weaker on both datasets and is therefore interpreted only as architectural robustness evidence.

Because each user has one held-out target, sampled-set HitRate@10 is numerically the same indicator as a one-target recall measure; HitRate is used here to avoid any implication of catalogue-level retrieval recall.

Table 3. Recommendation-quality and masking-gate audit on the held-out target plus the stated number of unseen negatives. HitRate@10 is the one-target hit indicator, not catalogue retrieval recall. Values are distinct-user means with 10,000-draw user-bootstrap 95% CIs; popularity references are per-dataset constants. Gate reports passing model seeds out of five. ML-1M = MovieLens-1M.

Dataset	Model	NDCG@10 [95% CI]	HR@10 [95% CI]	Pop. NDCG	Pop. HR	Gate
ML-1M	ItemKNN	0.284 [0.263, 0.304]	0.504 [0.473, 0.535]	0.158	0.320	Pass (5/5)
Amazon	ItemKNN	0.181 [0.161, 0.201]	0.285 [0.257, 0.314]	0.090	0.155	Pass (5/5)
ML-1M	Profile	0.156 [0.140, 0.172]	0.301 [0.275, 0.326]	0.158	0.320	Boundary (4/5)
Amazon	Profile	0.038 [0.033, 0.044]	0.073 [0.065, 0.082]	0.090	0.155	Pass (5/5)

6.2 Faithfulness versus bounded alignment

Table 4 separates deletion AIA, bounded AIA, and their difference. On MovieLens ItemKNN, Shapley has bounded AIA 0.779 versus 0.933 for LIME and 0.978 for LOO. On Amazon, Shapley has bounded AIA 0.414 versus 0.827 and 0.851. Thus Shapley does not win absolute alignment. This conclusion is estimator-robust: replacing the reverse-paired Monte Carlo estimator with KernelSHAP at LIME's mask budget (256/512 masks) leaves bounded alignment at 0.65–0.66 on MovieLens and 0.63 on Amazon, far below LIME and LOO in either case, and the two Shapley estimators agree strongly (Pearson 0.78–0.98; Supplementary Table S25).

The contrast is in the change between perturbations. Shapley's primary gap is +0.0169 on MovieLens and +0.1285 on Amazon. LIME and LOO are negative. The random control is close to zero and its confidence intervals include zero in the primary conditions. Greedy is negative on the primary conditions. The gap therefore measures a change of evaluation target; it is not an explanation-validity score.

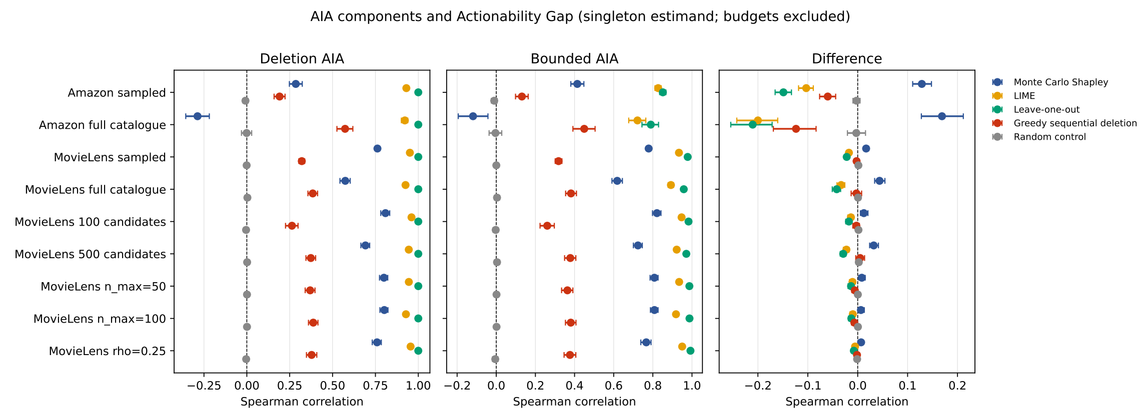

Fig. 2. Deletion AIA, bounded-intervention AIA, and their difference for the five methods. Points are distinct-user means and intervals are user-bootstrap 95% confidence intervals. Budget rows are excluded because budgets do not define singleton gap estimands. The underlying values are also supplied in editable Table 4 and the alignment table in Supplementary Section S7, so interpretation does not depend on color alone.

Table 4. Primary ItemKNN alignment components. Values are distinct-user means; bounded AIA and Gap include 95% user-bootstrap confidence intervals. Deletion-AIA intervals are provided in Supplementary Section S7. Shapley = Monte Carlo Shapley; Greedy = greedy sequential deletion; Random = random control.

Dataset	Method	Deletion AIA	Bounded AIA [95% CI]	Gap [95% CI]	Valid n
MovieLens	Shapley	0.762	0.779 [0.769, 0.789]	+0.017 [+0.013, +0.022]	1000
MovieLens	LIME	0.951	0.933 [0.928, 0.938]	−0.018 [−0.022, −0.014]	1000
MovieLens	LOO	1.000	0.978 [0.974, 0.981]	−0.022 [−0.026, −0.019]	1000
MovieLens	Greedy	0.321	0.318 [0.301, 0.336]	−0.002 [−0.007, +0.002]	1000
MovieLens	Random	−0.001	0.000 [−0.006, +0.007]	+0.001 [−0.001, +0.005]	1000
Amazon	Shapley	0.286	0.414 [0.381, 0.448]	+0.129 [+0.110, +0.148]	993
Amazon	LIME	0.930	0.827 [0.811, 0.843]	−0.103 [−0.119, −0.089]	993
Amazon	LOO	1.000	0.851 [0.835, 0.867]	−0.149 [−0.166, −0.133]	993
Amazon	Greedy	0.191	0.131 [0.098, 0.163]	−0.060 [−0.076, −0.044]	993
Amazon	Random	−0.008	−0.010 [−0.025, +0.005]	−0.002 [−0.010, +0.005]	993

6.3 Joint intervention decisions

The operational outcome is NDCG@10 under the exact $B = 2$ action space. On MovieLens ItemKNN, LOO yields mean NDCG effect 0.0431, LIME 0.0425, and Shapley 0.0403. Success is 28.9%, 28.8%, and 27.4%, respectively. Conditional normalized regret among 339 active-oracle users is 0.217, 0.225, and 0.268. Local methods therefore retain a modest decision advantage on MovieLens.

On Amazon ItemKNN, NDCG effects are 0.0333 for Shapley, 0.0331 for LIME, and 0.0313 for LOO. Conditional normalized regret among 196 active-oracle users is 0.206, 0.197, and 0.232. The differences are small and do not support a universal winner. Random actions have negative mean NDCG effects on both datasets. Signed alignment—the Spearman correlation between predicted benefits $-\phi^{g}_{u,p}$ and bounded effects $\Delta^{z,\mathrm{bnd}}_{u,p}$, distinct from direction accuracy (Eq. (17))—is high for the three principal explainers and much lower for the controls (diagnostics table in Supplementary Section S8:

Table 5. Primary budget-two intervention decisions. NRegret is conditional on active NDCG oracles; the active-user count is shown in parentheses.

Dataset	Method	ΔNDCG	Success	Abstention	NRegret (n)
MovieLens	Shapley	0.040	27.4%	0.0%	0.268 (339)
MovieLens	LIME	0.043	28.8%	0.0%	0.225 (339)
MovieLens	LOO	0.043	28.9%	0.0%	0.217 (339)
MovieLens	Greedy	0.006	10.3%	0.0%	1.471 (339)
MovieLens	Random	-0.001	7.4%	0.0%	1.446 (339)
Amazon	Shapley	0.033	17.0%	1.0%	0.206 (196)
Amazon	LIME	0.033	17.1%	7.0%	0.197 (196)
Amazon	LOO	0.031	16.7%	8.6%	0.232 (196)
Amazon	Greedy	0.010	9.0%	0.7%	0.784 (196)
Amazon	Random	-0.004	6.6%	4.8%	1.197 (196)

Amazon 0.855 Shapley, 0.949 LIME, 0.966 LOO, 0.109 Greedy, 0.009 Random; MovieLens 0.929, 0.974, 0.994, 0.342, 0.004), so the magnitude-alignment ordering transfers to sign validity.

For the Amazon Shapley-versus-LIME realized-NDCG comparison we declare a practical equivalence margin of ±0.005 ΔNDCG as the smallest decision-relevant effect size (approximately one eighth of the typical realized effect magnitude 0.04) and apply TOST on the seed-averaged paired user differences (n = 993 on Amazon after constant-vector exclusion): the mean difference is +0.00013 with 90% CI [−0.00162, +0.00188] entirely inside ±0.005 (both one-sided $p < 10^{-4}$), so the two methods are practically equivalent on Amazon realized NDCG under this margin; we therefore report equivalence within a declared margin rather than interpreting a nonsignificant test as indistinguishability. On MovieLens the same test also falls inside ±0.005 (mean −0.00227, 90% CI [−0.00365, −0.00088]) while excluding zero, i.e. LIME is detectably but practically slightly better. Normalized regret is right-skewed: medians are 0 for Shapley/LIME/LOO (fraction above one ≤ 0.3%), whereas Greedy reaches the 95th percentile 2.81 on MovieLens (15.9% of users above one) and Random 3.39 (33.0%), so mean regret differences are driven by a harmful-action tail that the signed, abstention-aware rule avoids.

Primary NDCG-effect 95% user-bootstrap CIs: MovieLens Shapley [.035, .046], LIME [.037, .049], LOO [.037, .050]; Amazon Shapley [.027, .040], LIME [.027, .039], LOO [.026, .037]. Individual-method conditional NRegret 95% CIs (active-oracle users): MovieLens Shapley [.227, .309], LIME [.187, .264], LOO [.178, .258], Greedy [1.114, 1.895], Random [1.227, 1.698] (n = 339); Amazon Shapley [.158, .256], LIME [.150, .248], LOO [.180, .287], Greedy [.639, .954], Random [1.052, 1.369] (n = 196). Active-oracle fractions are 339/1000 (MovieLens) and 196/993 (Amazon); pairwise NRegret differences with Holm-adjusted p-values are in Table S5.

6.4 Hypothesis adjudication

The two declared hypotheses are adjudicated here against the primary results. **H1** (deletion and bounded interventions probe distinct, method-specific response surfaces) is supported: the bounded-minus-deletion gaps in Table 4 are non-zero and method-specific—positive for Shapley (+0.017 MovieLens, +0.129 Amazon, bootstrap CIs excluding zero), negative for LIME (−0.018, −0.103) and LOO (−0.022, −0.149)—so the scores shift materially with perturbation semantics. **H2** (pointwise alignment does not determine decision quality) is also supported: LIME and LOO dominate Shapley on absolute bounded alignment, yet the realized-NDCG differences among the three methods are small, dataset-inconsistent, and—on Amazon—within the declared equivalence margin (Section 6). The declared falsification check asks whether the

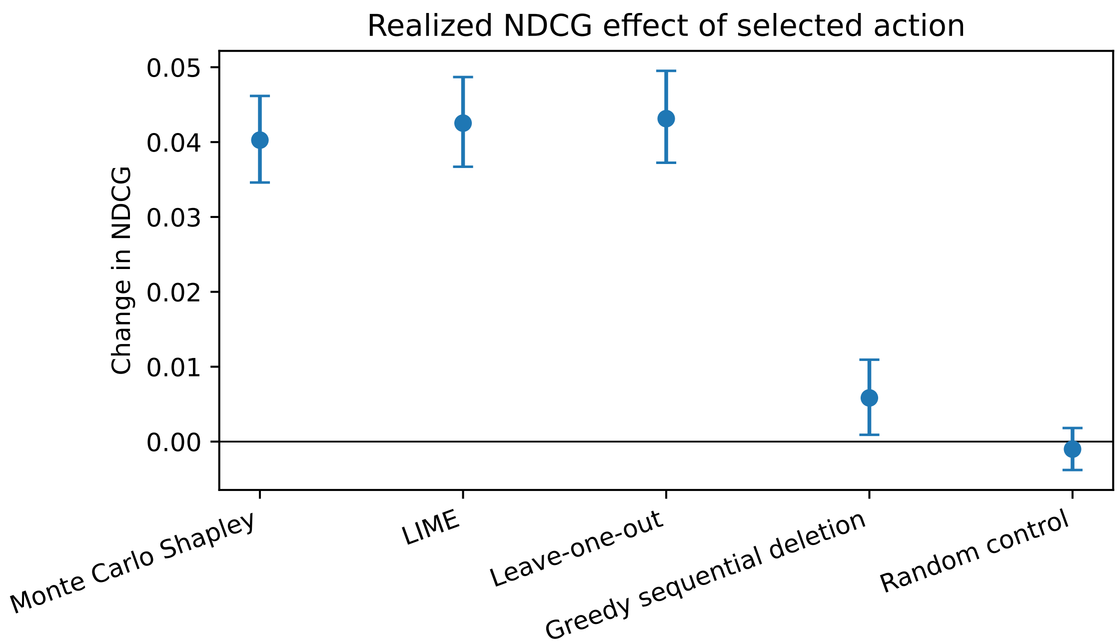

Fig. 3. Primary MovieLens ItemKNN NDCG effect for actions selected at budget two. We show MovieLens for visual legibility; the analogous Amazon effects are reported in Table 5. The comparison is retrospective and target-conditioned; intervals are distinct-user bootstrap intervals.

gap predicts decision quality: it does not. Across users, the Spearman correlation between the bounded-minus-deletion gap and normalized regret is +0.035 ($p = 0.27$, $n = 1000$) on MovieLens and +0.042 ($p = 0.19$, $n = 987$) on Amazon for Shapley, and the correlation with realized NDCG effect is −0.054 ($p = 0.085$) and −0.097 ($p = 0.002$), respectively. These results do not support reading a positive gap as decision-relevant, which is why the gap is reported only as a perturbation-sensitivity diagnostic.

6.5 Null calibration and convergence

Table 6 reports the complete within-user null calibration alongside the component table. The within-user AIA null is centred near zero for all primary ItemKNN conditions, and Shapley, LIME, LOO, and greedy sequential deletion exceed the null for bounded magnitude AIA, while random attribution does not. This establishes that bounded alignment is not merely a consequence of the correlation statistic returning positive values.

Exact enumeration is feasible only for short profiles: the 250-user replication runs enumerate 89/87 users with $n_u \leq 12$ on Amazon (ItemKNN/SASRec) and 6/8 on MovieLens, whose histories are longer, while a dedicated validation on the 1,000-user primary cohort enumerates 300 Amazon and 73 MovieLens users (Supplementary Section S9); the two counts refer to these different cohorts. Both show model- and dataset-dependent agreement of the reverse-paired estimator: MovieLens ItemKNN reaches a mean per-user maximum absolute error of 2.1×10^{-3} with mean rank agreement $\rho = 0.982$, whereas Amazon ItemKNN reaches 1.1×10^{-4} but only $\rho = 0.783$; under SASRec the corresponding mean agreements are $\rho = 0.688$ (MovieLens) and $\rho = 0.395$ (Amazon). A small absolute error therefore does not imply

Table 6. Primary within-user null calibration of bounded target-margin AIA. Observed values are bounded-intervention magnitude AIA; the $R_{\text{null}} = 1{,}000$ matched null draws and observed values are averaged across the five seeds within each distinct user before summarisation. p-values are unadjusted upper-tail null p (Eq. 20), i.e. descriptive calibration controls, not a confirmatory family. For compactness, null means are shown in 10^{-4} units while observed AIA and null 95th percentiles are shown in raw AIA units.

Dataset	Method	AIA	Null mean (10^{-4})	Null p95	Unadj. upper-tail null p	n
MovieLens	Shapley	0.7786	−0.8	0.0055	0.0010	1000
MovieLens	LIME	0.9334	+0.8	0.0059	0.0010	1000
MovieLens	LOO	0.9780	−1.1	0.0059	0.0010	1000
MovieLens	Greedy seq. del.	0.3183	+2.4	0.0061	0.0010	1000
MovieLens	Random	0.0003	0.0	0.0058	0.4635	1000
Amazon	Shapley	0.4141	−0.3	0.0128	0.0010	993
Amazon	LIME	0.8271	−3.7	0.0128	0.0010	993
Amazon	LOO	0.8511	+0.7	0.0126	0.0010	993
Amazon	Greedy seq. del.	0.1310	−2.4	0.0124	0.0010	993
Amazon	Random	−0.0103	+0.7	0.0125	0.9091	993

Table 7. Selected target-margin Monte Carlo convergence budgets. Only selected rows are shown. M_{pair} counts base permutations and T counts evaluated forward/reverse orders.

Dataset	Model	M_{pair}	T	Rank	Jaccard	Rank-valid	Coverage
Amazon	itemknn	50	100	0.9532	0.9090	100.0%	60.0%
Amazon	profile	100	200	0.9670	0.9110	100.0%	66.0%
MovieLens	itemknn	250	500	0.9853	0.8433	100.0%	69.0%
MovieLens	profile	50	100	0.9670	0.8320	100.0%	56.0%

high rank agreement when the exact attribution spectrum is compressed or nearly tied, and Monte Carlo accuracy is validated separately per dataset and architecture.

Under the population-average diagnostic criteria, target-margin convergence selects $M_{\text{pair}} = 50$ for Amazon ItemKNN, $M_{\text{pair}} = 100$ for Amazon profile, $M_{\text{pair}} = 250$ for MovieLens ItemKNN, and $M_{\text{pair}} = 50$ for MovieLens profile, with per-user joint-threshold coverage at the selected budgets of 0.56–0.69 (rising to at most 0.89 at $M_{\text{pair}} = 1000$, and 0.74–0.85 at the 500–1000 floor in the quantile table of Supplementary Section S8); these settings are therefore mean-converged and are conservatively overridden by the final floor $M_{\text{pair}} = 500$ ($T = 1000$ evaluated orders). Rank-valid fraction is the fraction of users with finite rank comparisons, whereas threshold coverage is the fraction of rank-valid users satisfying both the rank and action-overlap thresholds; coverage is diagnostic and is not a replacement for the rank-validity criterion. The NDCG attribution sensitivity remains unconverged at $M_{\text{pair}} = 1000$ and is reported only as a stress test.

6.6 Robustness and boundaries

Before reporting the sensitivities we state their interpretive ceiling. Scaling the Amazon full-unseen-catalogue audit from 250 to the full 1,000-user cohort (median catalogue 8,572 unseen items; Table 8) raises the number of active target-margin oracles to 883. At catalogue scale, LIME and leave-one-out retain positive bounded alignment (0.71 and 0.78) while Monte Carlo Shapley is mildly anti-aligned (−0.050) despite a positive bounded-minus-deletion difference (+0.16); the difference therefore again reflects reduced anti-alignment rather than successful bounded-intervention prediction. Full-catalogue decision effects are near zero for every method, so catalogue-scale decision conclusions remain descriptive. With that caveat, the bounded-minus-deletion difference for Shapley stays positive across the primary ItemKNN candidate, history, and intervention-strength conditions, while absolute values vary. Full-catalogue NDCG alignment is sparse (54 active NDCG oracles) and is treated descriptively. The longer-history conditions also

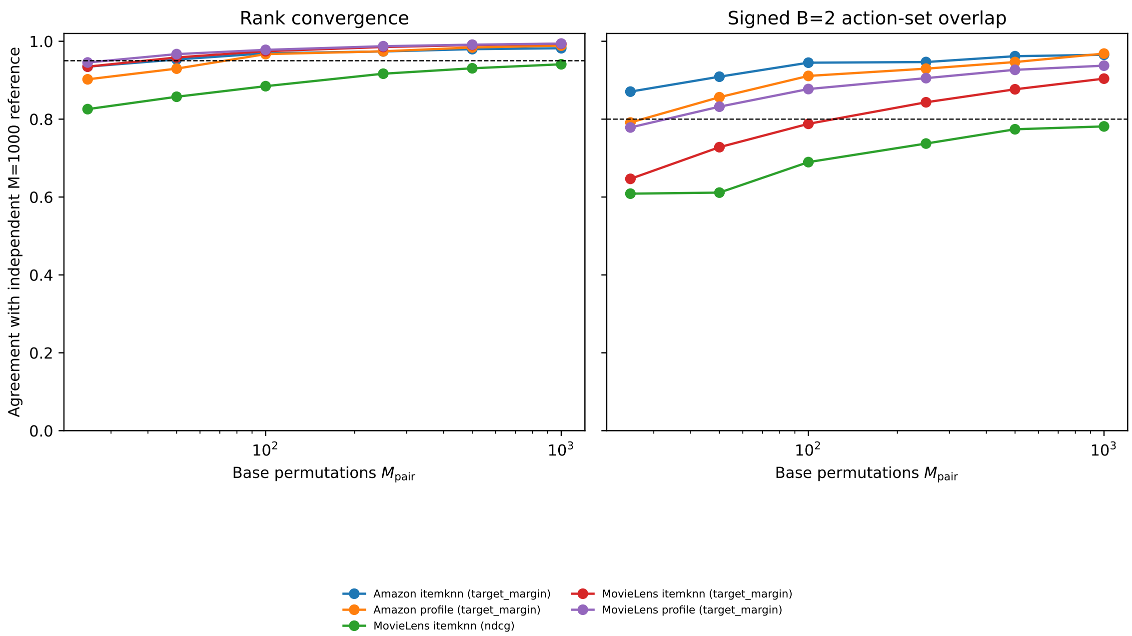

Fig. 4. Independent Monte Carlo convergence against an independent $M_{\text{pair}} = 1000$ *MC reference* (not ground truth; an exact-reference subset is specified in the replication guide). Left: rank agreement with the reference; right: signed $B = 2$ action-set Jaccard overlap, the second selection criterion; dashed lines mark the 0.95 and 0.80 thresholds. The legend maps every curve to its dataset–model–utility condition. The numerical values are provided in Supplementary Table S6, so interpretation does not rely on colour alone. Efficiency is not used as a convergence criterion because the prefix walk telescopes.

show reduced masking sensitivity, supporting the use of $n_{\max} = 20$ as a declared operating boundary rather than an arbitrary computational cap.

Table 8. Amazon Digital Music full-unseen-catalogue audit at 1,000 users (five seeds, seed-mean within user; median catalogue size 8,572 unseen items). Bounded and deletion AIA, their difference, signed alignment, realized target-margin effect of the selected action, and conditional normalized regret; intervals are user-bootstrap 95% CIs. Active-oracle denominators: 883 (target margin) and 54 (NDCG).

Method	Bounded AIA	Deletion AIA	Gap	Signed	Δ effect	Norm. regret
Monte Carlo Shapley	-0.05 [-0.09,-0.01]	-0.21 [-0.25,-0.17]	0.16 [0.14,0.18]	0.71 [0.69,0.73]	0.0015 [0.0013,0.0017]	0.74 [0.53,1.04]
LIME	0.71 [0.69,0.73]	0.92 [0.91,0.92]	-0.21 [-0.23,-0.19]	0.93 [0.92,0.94]	0.0015 [0.0014,0.0017]	0.43 [0.40,0.46]
Leave-one-out	0.78 [0.76,0.80]	1.00 [1.00,1.00]	-0.22 [-0.24,-0.20]	0.95 [0.94,0.96]	0.0016 [0.0014,0.0017]	0.41 [0.38,0.44]
Greedy sequential deletion	0.42 [0.39,0.44]	0.55 [0.52,0.57]	-0.13 [-0.15,-0.11]	0.08 [0.05,0.11]	-0.0009 [-0.0010,-0.0007]	2.97 [2.16,4.29]
Random control	-0.01 [-0.02,0.01]	-0.00 [-0.02,0.01]	-0.00 [-0.01,0.01]	-0.00 [-0.02,0.01]	-0.0011 [-0.0012,-0.0010]	2.59 [2.15,3.16]

The profile model supplies a useful negative boundary: it can change rankings while underperforming popularity, and one MovieLens seed fails the NDCG masking threshold. This demonstrates why a non-degenerate masking gate and recommendation-quality audit are necessary before interpreting attribution results.

History length explains part, but not all, of the cross-dataset contrast (Table 10). The Amazon Shapley difference remains positive in every history stratum, including $n_u = 20$; MovieLens is also positive, although its dominant $n_u = 20$ stratum has a smaller mean difference. The nine-user MovieLens 3–5 stratum is too small for a stable cross-dataset

Table 9. Paired seed-mean differences on the same 1,000 full-catalogue users (user-bootstrap 95% CIs).

Left	Right	Metric	n	Diff.	95% CI
Monte Carlo Shapley	LIME	bounded AIA	999	-0.760	[-0.809,-0.710]
Monte Carlo Shapley	LIME	Δ effect	1000	-0.000	[-0.000,0.000]
Monte Carlo Shapley	Leave-one-out	bounded AIA	999	-0.828	[-0.876,-0.780]
Monte Carlo Shapley	Leave-one-out	Δ effect	1000	-0.000	[-0.000,0.000]
LIME	Leave-one-out	bounded AIA	999	-0.068	[-0.078,-0.059]
LIME	Leave-one-out	Δ effect	1000	-0.000	[-0.000,-0.000]

Table 10. History-length-stratified Shapley magnitude-alignment difference. Intervals are descriptive 95% user-bootstrap intervals; small MovieLens strata should be interpreted cautiously.

Dataset	Players n_u	Valid n	Gap [95% CI]
Amazon	3–5	580	.121 [.094,.150]
Amazon	6–10	262	.167 [.137,.199]
Amazon	11–19	92	.099 [.063,.138]
Amazon	20	59	.076 [.048,.109]
MovieLens	3–5	9	.373 [.095,.722]
MovieLens	6–10	42	.028 [.006,.052]
MovieLens	11–19	118	.012 [.001,.023]
MovieLens	20	831	.013 [.010,.016]

comparison and is flagged in the table caption. Accordingly, "especially on Amazon" is reported as a descriptive empirical pattern rather than a dataset-causal conclusion.

A replication of selected audit components under a nonlinear sequential scorer (a SASRec-style transformer) is reported in Supplementary Section S6, together with the intervention-aware (path-matched) baselines and the prospective, non-target-conditioned audits. Its headline findings are that path-matched baselines match or exceed Monte Carlo Shapley when pair interactions are strong, that prospective audits leave the alignment conclusions intact, and that the signed, abstention-aware selection rule is supported by the component ablation. Beyond this, the central alignment finding generalizes: path-matched baselines (finite differences, integrated gradients, bounded LIME) exceed Monte Carlo Shapley on bounded-intervention alignment under a graph architecture (LightGCN) and on a third dataset from a different domain (Gowalla location check-ins), as detailed in Supplementary Section S9.

7 Discussion and Broader Implications

The results support a three-axis interpretation. LIME and LOO are strongest on absolute pointwise alignment because their constructions are local to deletion or the full profile. Shapley has lower absolute alignment but a larger positive change when the evaluation moves from deletion to bounded downweighting. This is consistent with coalition-context averaging, but the experiment does not establish a causal mechanism and the random control prevents interpreting the gap as standalone validity.

Table 11 compares the protocol dimensions most relevant to the closest counterfactual and explanation-evaluation studies. The seven criteria—player type, intervention policy, outcome, joint oracle, inference unit, bounded/continuous support, and null control—determine whether a protocol can audit attributions against feasible, budgeted joint decisions with statistically valid user-level evidence.

ActionShap is an evaluation protocol, not a counterfactual generator or attribution algorithm: every explainer is audited against the same declared bounded intervention policy and exact action space, after which singleton alignment

Table 11. Component-level positioning against the closest related evaluation and counterfactual recommendation work.

Work	Players	Policy	Outcome	Oracle	Inference	Bounded/cont.	Null ctrl.
PRINCE [21]	Provider, profile	Counterfactual gen.	Rec. change	Method-spec.	Method-spec.	No	No
ACCENT [71]	User factors	Counterfactual masking	Top-K expl.	None	Method-spec.	No	No
Tan et al. [70]	Rec. factors	Counterfactual expl.	Rank change	Method-spec.	User/item	No	No
CEERS [4]	Expl. factors	Counterfactual eval.	Expl. quality	Benchmark	Repro.	No	No
Barkan [6]	Learned masks	Model-spec. cf.	Expl. quality	Learned	Method-spec.	Learned	No
LXR [24]	Learned masks	Model-spec. cf.	Expl. quality	Learned	Method-spec.	Learned	No
Mohammadi [47]	Expl. factors	Top-K cf. eval.	Consistency	None	Benchmark	No	No
Nguyen [51]	User/profile	Top-K cf. eval.	Effect., sparsity	Benchmark	Cross-paper	No	No
RankSHAP [14]	Rank features	Attribution	Rank fidelity	None	Feature-level	No	No
ShaRP [57]	Rank elements	Attribution	Rank preference	None	Feature-level	No	No
TimeSHAP [9]	Sequence events	Event deletion	Sequence output	None	Feature-level	No	No
Ref. fidelity [5]	Expl. factors	Cf./deletion	Fidelity	Benchmark	Benchmark	No	No
CLIME [60]	Features	Deletion	Local fidelity	None	Feature-level	No	No
Session attr. [10]	Session events	Deletion	Session ranking	None	Feature-level	No	No
ActionShap	History interactions	Bounded downweight	AIA, NDCG, regret	Exact $B = 2$; abstain	Distinct-user	Binary+cont.	Within-user

and joint decision regret are measured with abstention permitted and inference performed over distinct users with an explicit chance control. Relative to the closest work (Table 11), including recent counterfactual-explanation benchmarks and studies showing that recommender performance itself alters explainer evaluation [46, 51], the non-overlap is the combination of bounded intervention semantics, null-calibrated attribution alignment, and exact joint-action regret under a common feasible space.

The decision results reinforce this distinction. On MovieLens, local methods select slightly better NDCG actions. On Amazon, Shapley and LIME are close, with modest differences in conditional regret. A method can therefore show a positive change between deletion and bounded-intervention alignment without selecting the best action for every operational utility. This is precisely why ActionShap reports pointwise alignment, signed direction, success, abstention, and regret together.

When should a practitioner use this protocol instead of a deletion audit? Consider a platform that can discount suspect historical interactions—suspected bot clicks, stale preferences, or disputed ratings—but cannot delete them, so the executable policy is a bounded downweighting at some operating strength. A deletion audit ranks interactions by their removal effect; if the operational policy is bounded, that ranking can misestimate which interactions actually move the recommendation, because deletion and bounded downweighting probe different response surfaces (Section 6.6 shows methods re-ranking between the two semantics). ActionShap answers three concrete questions a deletion audit cannot: (i) does this attribution predict bounded intervention effects at the operating strength (bounded AIA and signed directional accuracy); (ii) would acting on it select a near-oracle joint action under the declared budget (regret against the exact oracle); and (iii) are these conclusions distinguishable from chance for this user population (within-user null calibration)? A positive bounded-minus-deletion gap is not itself a decision criterion—the results show it does not select better actions—but it flags exactly the settings in which deletion-based conclusions are fragile, which is when a bounded audit is worth running. The protocol also clarifies what offline actionability means. The intervention is system-executable inside the frozen simulator: it changes an inference-time interaction weight without retraining. It is not automatically a user action, causal intervention, or prospective policy. A production deployment would need authority, cost, user-consent, and feedback-loop specifications that are outside this benchmark.

From a responsible-AI standpoint, ActionShap is an oversight instrument rather than a deployment policy: it makes explicit which attributions correspond to feasible, predictable changes under a declared policy, and it treats a positive bounded-minus-deletion difference as descriptive rather than as evidence of user-facing actionability. This supports transparency and human oversight without claiming perceived usefulness, which remains outside the benchmark's scope.

7.1 Limitations and Threats to Validity

7.1.1 Construct validity. Actionability is defined as predictive validity under one bounded downweighting policy. It is not perceived usefulness, legal recourse, user agency, or a causal effect in the world. All primary players are technically mutable, so feasibility does not differentiate factors; future work should study heterogeneous costs and user-side constraints.

7.1.2 Metric validity. NDCG is discrete and produces constant effect vectors for some users. We report missing correlations rather than imputing zero. Target margin is the converged attribution utility but has less direct operational meaning; NDCG is the action outcome and is never relabelled as target margin.

7.1.3 Internal validity. Temporal splitting and complete pre-test exclusion reduce leakage, but the audit remains target-conditioned. The held-out target defines the labelled instance being explained, which is appropriate for retrospective evaluation but unavailable to a prospective policy. Candidate sets are fixed and target coverage is one by construction; sampled ranking is not retrieval evaluation.

7.1.4 External validity. Including the supplementary replications, the empirical base spans three datasets (MovieLens-1M, Amazon Digital Music, and Gowalla) and three scorer families (ItemKNN, a SASRec-style sequential transformer, and LightGCN), and the central alignment finding replicates across them. Multimodal inputs, online feedback, provider objectives, and heterogeneous intervention costs nevertheless remain uncovered. The SASRec-style and LightGCN cells both underperform popularity under the protocol's inference-time history-weighting interface, so they demonstrate applicability rather than transfer to a competitive neural recommender. A tuned LightGCN variant (deeper propagation, larger embeddings, higher learning rate, and 80 epochs) also remains below popularity, suggesting that the scoring-time weighting interface, rather than the tested training budget, remains a limiting factor (Supplementary Section S9). The full-catalogue subset is smaller than the primary cohort, and its active-oracle counts are low; scaling full-catalogue evaluation and replacing the seeded Bernoulli LIME mask design with enumeration (feasible at Amazon's short histories) or without-replacement sampling remain validation extensions. The profile model's quality shortfall and one failed gate are retained as explicit boundaries rather than generalized.

Three further scope limitations are declared. First, each user is audited on a single held-out target (a one-shot retrospective audit); stability across alternative targets is not measured. Second, the estimand and analysis plan appear in the frozen experimental contract and this manuscript, but no external preregistration record exists. Third, both primary datasets are entertainment/e-commerce implicit-feedback domains in English-language markets, and the Amazon Digital Music cohort is reconstructed from the public Amazon Reviews 2018 release by the documented thresholding and 5-core procedure; the recorded source and processed hashes identify our reconstruction, which may differ in detail from other published variants.

7.1.5 Selective reporting and selection threats. Every table and figure was generated from the schema-v2 results by a validator whose build fails on protocol violations. The validation report and machine-readable matrices will accompany the public archive; until deposit, the manuscript package preserves the generated assets but does not independently regenerate them. Evaluation users, candidates, tie priorities, model seeds, and attribution seeds are fixed, distinct, and seeded rather than ordered by identifier, which removes identifier-order selection effects; the same users and candidate sets are reused across methods, seeds, and sensitivities. Hardware-specific effects are limited to floating-point determinism; exact dependency versions are recorded, and no hardware-dependent quantity (processor model, RAM, peak RSS) is claimed because it was not serialized.

7.1.6 Statistical conclusion validity. The discrete NDCG outcome yields constant effect vectors for many users, so normalized regret is conditional on the 339/196 active-oracle users and success rates are low by construction; these denominators are reported with every decision statistic, and full-catalogue decision cells with single-digit active counts are labelled exploratory. The NDCG attribution game did not meet the frozen convergence threshold at $M_{\text{pair}} = 1000$ and is retained only as a descriptive stress test, never as a primary estimand. Three further conditioning choices bound the inference: (i) all bootstrap and permutation intervals are conditional on one frozen trained recommender per model—users share the learned similarity structure, so intervals describe variability across sampled evaluation users given the fitted model and cohort, not training-distribution uncertainty; (ii) the confirmatory multiplicity families are the dataset–model–condition–metric groups of the paired-test release, each containing the ten declared method pairs, corrected by Holm within family; and (iii) equivalence is claimed only where a declared-margin TOST rejects both one-sided nulls (Section 6); nonsignificant pairwise tests are otherwise reported as no detected difference.

8 Conclusion

ActionShap provides a fully specified protocol for asking whether recommendation explanations predict the effects of declared system-executable profile interventions. In the primary ItemKNN experiments, Shapley exhibits a positive bounded-minus-deletion alignment difference, particularly on Amazon, while LIME and LOO retain higher absolute alignment and slightly better MovieLens NDCG decisions. The random control remains near its chance null, and a positive bounded-minus-deletion difference is therefore not sufficient evidence of explanation validity. The principal contribution is the separation of deletion faithfulness, bounded-intervention alignment, and joint decision quality under a common, leakage-safe, abstention-aware protocol. Future work should extend the audit to competitive neural and graph recommenders that both pass the quality gate and expose the bounded-weight interface, datasets beyond entertainment, user studies of decision support, and heterogeneous intervention costs.

9 Data Availability

MovieLens-1M was obtained from the GroupLens public release, and Amazon Digital Music was reconstructed from the Amazon Reviews 2018 public release after the documented rating threshold, deterministic deduplication, and iterative 5-core filtering. Source and processed-data SHA-256 values, split rules, and candidate construction are recorded in the provenance manifest. The datasets remain subject to their maintainers' terms and are not redistributed with the artifact.

10 Code and Materials Availability

The submission package includes the editable LaTeX tables and vector figures used in the manuscript. The accompanying artifact will contain the generation code, configurations, preprocessing and execution scripts, validation reports, provenance manifest, and machine-readable result matrices. **The permanent or reviewer-view artifact URL must be inserted here before submission.**

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AI-assisted preparation. The authors used OpenAI Codex for LaTeX formatting, consistency checking, and language editing. The authors reviewed and edited all resulting material and take full responsibility for the content of the work.

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A Formal Guarantees

A.1 Efficiency of the prefix-walk estimator

For any permutation $\pi = (p_1, \ldots, p_n)$, define $S_0 = \varnothing$ and $S_j = \{p_1, \ldots, p_j\}$. The prefix-walk contribution of p_j is $v(S_j) - v(S_{j-1})$. Consequently,

$$\sum_{j=1}^{n} \left[v(S_j) - v(S_{j-1}) \right] = v(P_u) - v(\varnothing). \tag{A.1}$$

Averaging over paired permutations preserves the identity up to floating-point arithmetic. The efficiency residual in the release is therefore a cache and arithmetic diagnostic, not a convergence criterion.

A.2 Leave-one-out ceiling

For deletion at $\rho = 0$, $\Delta^{\text{del}}_{u,p} = v(P_u \setminus \{p\}) - v(P_u)$ and $\phi^{\text{LOO}}_{u,p} = v(P_u) - v(P_u \setminus \{p\})$. Thus $|\phi^{\text{LOO}}_{u,p}| = |\Delta^{\text{del}}_{u,p}|$ for every valid nonconstant user and deletion AIA is one. LOO's deletion-to-bounded gap is consequently bounded above by zero; it is retained as a diagnostic oracle, not a positive-gap competitor.

A.3 Exact budget-two action space

With $n_u \le 20$, the exact action space contains $1 + n_u + \binom{n_u}{2} \le 211$ actions, including no action. Each action is evaluated against target margin and NDCG using the same cached candidate scores. The oracle is therefore exact for every primary user and does not rely on a sampled action subset.

B Experimental Contract

The profile optimizer samples one positive item per user per epoch, excludes that positive from its temporary context, and samples a negative uniformly outside the user's complete training history. ItemKNN has no latent user embedding, shrinkage coefficient, minimum-support cutoff, or negative-similarity branch: the retained similarity matrix is non-negative by construction. AIA is Spearman rank correlation and is missing for constant or non-finite vectors. Normalized regret uses $\varepsilon = 10^{-12}$ and is averaged only over positive-oracle users. The artifact records the exact dependency versions,

Table B.1. Frozen experimental contract used for every paper-eligible run. The executable YAML and scripts are the source of truth; this table records every parameter that changes the estimand or reported uncertainty.

Element	Primary setting	Robustness/sensitivity settings
Datasets	MovieLens-1M; Amazon Digital Music, rating ≥ 4	same
Temporal split	last test, penultimate validation; timestamp/index ties	complete pre-test exclusion
Model fitting	complete training histories	no retraining under intervention
Player/scoring history	most recent $n_{\max} = 20$; older B_u not scored	$n_{\max} \in \{50, 100\}$
ItemKNN	cosine co-occurrence; top 200 non-zero neighbours; no shrinkage	same
Profile model	64 dimensions; 10 BPR-style epochs; $\eta = .03$; $L_2 = 10^{-4}$	robustness-only
Candidates	target + 199 uniform unseen negatives	100, 500, full unseen catalogue
Tie, candidate, user seeds	31415/1729/2718	fixed across methods and seeds
Attribution utility	target margin, $L = 10$, unit-temperature sigmoid	NDCG stress test
Shapley sampling	$M_{\text{pair}} = 500$, $T = 1000$ reverse-paired orders	convergence sweep $M_{\text{pair}} = 25$–1000
LIME	512 masks; Hamming kernel width .25; ridge $\alpha = 1$	same
Intervention	simultaneous downweight $w_p \leftarrow .5w_p$	deletion and $\rho = .25$
Budget/oracle	exact all actions up to $B = 2$, including no action	$B = 1$ and $B = 3$ joint decisions
Null/inference	$R_{\text{null}} = 1000$ matched null draws; 10,000 bootstrap/permutation draws	Holm correction

source hashes, run durations, and selected users; complete per-user vectors remain in the compressed machine-readable asset rather than in the PDF.

C Notation Summary

Symbols with multiple roles are disambiguated by context: ρ denotes the intervention strength in $\mathcal{I}_\rho$ and ρ_S denotes Spearman correlation; B denotes the action budget, B_u the older (non-player) training context, and $\widehat{B}_u^g$ the predicted-benefit score. Players $p \in P_u$ (distinct interaction records); weights $w \in [0,1]^{P_u}$; embeddings $q_i, r_{u,-i} \in \mathbb{R}^d$; scores $s_i^{S,w}, f_u \in \mathbb{R}$; utilities $v_u^{\text{attr}}, q_u \in \mathbb{R}$ with $v_u^{\text{attr}}(\varnothing) = 0.5$; effects $\Delta_u^z \in \mathbb{R}$; τ global tie priority; $R_{\text{seed}} = 5$, $R_{\text{null}} = 10^3$, $R_{\text{boot}} = R_{\text{perm}} = 10^4$; V_u valid unordered seed pairs; $C_{\text{score}}(m, d)$ cost of scoring m candidates.

Supplementary Sections S7–S10 provide the robustness matrices, statistical validation and convergence diagnostics, replication summaries, and computational-cost budgets.